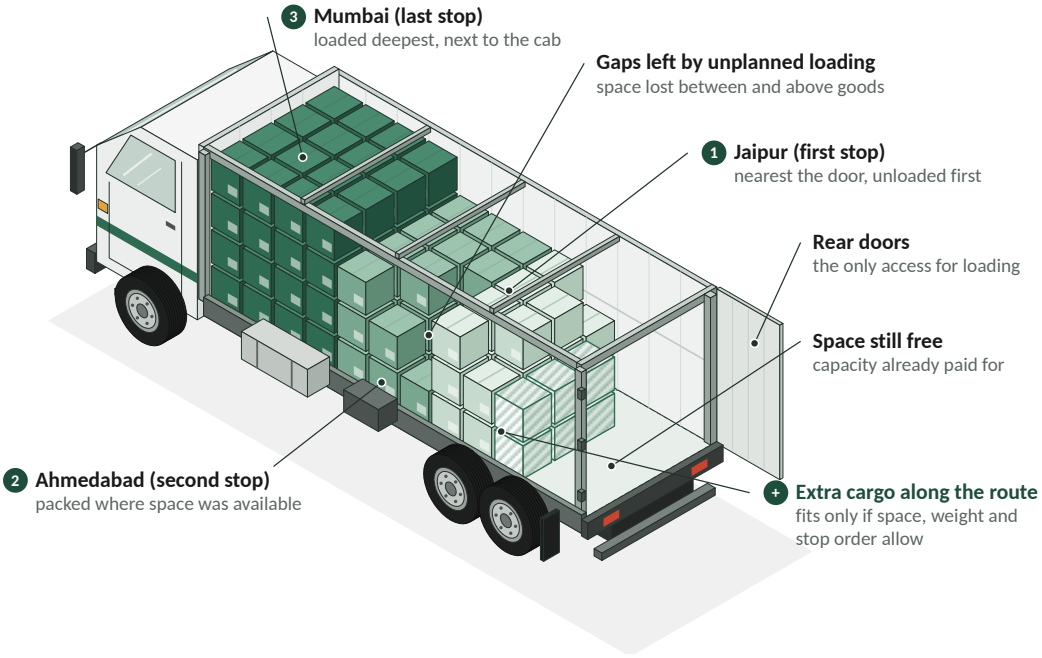
Spareway

Fuller trucks, smarter loading and better routes for India's truck operators

Team Bravo
Logistics & Supply Chain



THE PROBLEM IN BRIEF

Trucks across India regularly travel with part of their cargo space unused. As an illustration, if a truck with space for 100 air-conditioner units leaves with only 80, it still bears nearly the full cost of the journey while the remaining space earns nothing. Part of this capacity is lost when cargo is loaded without a clear layout, and part because operators have no practical way of finding and fitting additional cargo along the route they are already travelling.

The result is lower earnings for operators and drivers, more trips for the same amount of goods, overloaded or poorly arranged vehicles, and long, tiring journeys that are planned largely by habit.

1. Industry Space

Our project falls within **Logistics & Supply Chain**, with a focus on **road freight carried by small truck operators in India**. Road transport moves the majority of the country's domestic freight, and most of it is handled by owner-operators and small fleets rather than large logistics companies.

Our area of investigation is how efficiently a truck's capacity is used over the course of a trip: how cargo is arranged inside the vehicle, whether unused space can be filled with additional consignments along the way, and how the route itself is chosen. This area relates directly to two themes identified in the participant guide: the **waste of time and resources**, seen in unused space, repeated handling and unnecessary trips; and **poor coordination between the people involved**, as operators, drivers, loaders and customers each hold part of the information but rarely share it before the truck leaves.

2. Problem Statement

> **Small truck operators in India routinely run trips with part of their cargo capacity unused, because loads are arranged without a plan, additional cargo along the route is difficult to find and fit into an existing load, and routes are chosen without weighing cost, time and practicality together. This reduces earnings, increases the number of trips needed, and adds to the physical strain and risk of life on the road.**

2.1 Unused capacity on trips already being made

Every trip carries a fixed set of costs: fuel, tolls, the driver's wages and allowances, and wear on the vehicle. These costs are largely the same whether the truck is full or partly empty. When a truck that can carry 100 units leaves with 80, the space for the remaining 20 still travels the full distance at the operator's expense without producing any income. The same is true of return journeys that run empty or lightly loaded. National estimates suggest that **28–43% of truck travel in India is empty running**.[1]

2.2 Cargo arranged without a plan

Cargo is often loaded under time pressure, making it difficult to plan the placement of every item before a truck departs. Without a clear loading layout, gaps may remain between goods, available space may be poorly utilised, and cargo for earlier delivery stops may become blocked behind items scheduled for later destinations. This can require additional unloading and reloading, increasing handling time and the risk of damage.

The extent of space lost through loading practices can vary with the truck's dimensions, cargo type, packaging, stacking restrictions and delivery sequence. Spareway will investigate these factors through operator interviews and practical validation rather than assume a fixed percentage of wasted space. Its proposed 3D loading planner will help operators visualise available space and organise cargo before and during loading.

2.3 No practical way to add cargo along the route

Additional consignments are often available in towns and cities along a truck's route, but operators usually learn of them through brokers and phone calls, if at all. Even when an offer arrives, the operator cannot easily tell whether the new cargo will fit alongside the existing load, stay within the payload limit, and be unloaded without disturbing earlier drops. The decision is made in minutes and on estimates, so profitable cargo is either turned down or accepted at the risk of overloading and delay.

2.4 Routes chosen by habit, not by full cost

The route a truck takes determines its fuel use, toll charges, travel time, waiting at checkpoints and city entry points, and where the driver can eat and rest. These factors are rarely considered together. Routes are typically chosen by habit or on the advice of other drivers, and general navigation tools are designed mainly for shortest travel time rather than the overall cost and practicality of a freight journey.

2.5 Evidence that the problem exists

Evidence	What it indicates	Source
28–43% of truck travel is empty running	A large share of capacity on the road produces no income	NITI Aayog, 2021[1]
Typical load factors above 100% by weight	Trucks that are loaded are often loaded beyond safe limits, a sign of poor load decisions	NITI Aayog, 2021[1]
About 75% of the market run by operators with up to five trucks	The people most affected have the least access to planning tools	NITI Aayog, 2021[1]
Overloading linked to 12% of road crash deaths; about 49% of drivers caught with protruding loads reported paying a bribe	Poorly arranged and excess loads carry safety and financial costs	SaveLIFE Foundation, 2020[3]

2.6 Effects in the real world

Lost earnings

- Paid-for space travels empty
- Extra cargo is turned down
- More trips for the same goods

Safety risks

- Overloaded vehicles
- Unstable stacking
- Fatigue from long, unplanned journeys

Delays and damage

- Cargo reshuffled at stops
- Goods damaged in handling
- Late deliveries

Wider costs

- Higher fuel use per tonne carried
- Penalties and informal payments
- More congestion and emissions

3. Target User

Spareway is designed for **truck operators and the people who work on and around their trucks every day**: owner-operators and small fleet owners running one to five vehicles, the drivers who spend days at a time on the road, and the loaders and helpers who pack and unload the cargo. Our initial focus is closed-body trucks carrying packaged goods on intercity routes.

User	Role on a trip	How the problem affects them	What they need
Owner-operators and small fleet owners	Take bookings, decide what cargo to carry and set the route	Lose income on unused space and miss cargo they cannot evaluate in time	Clear visibility of spare capacity, nearby cargo and trip costs
Drivers	Drive the truck, often for many days, and handle drops along the way	Long hours, unplanned halts, delays when cargo has to be reshuffled	Practical routes, planned rest stops and an orderly load
Loaders and helpers	Load and unload cargo at each point	Repeated handling, physical strain, blame for damage and wasted space	A simple, step-by-step loading plan they can follow

3.1 How they experience the problem

Loading usually begins as soon as cargo reaches the truck. Loaders stack goods wherever there is room, the driver sets off once the booked cargo is on board, and the remaining space is often not measured. If another consignment becomes available on the way, the operator must judge by memory whether it will fit and whether the detour is worthwhile. The route follows familiar roads, and halts are taken wherever the driver happens to be when fatigue sets in. Taken together, these practices leave capacity unused and make journeys longer and more tiring than necessary.

3.2 Designed for the people who will use it

Many of our users are expected to work long hours, use low-cost Android phones and be more comfortable in Hindi or a regional language than in English. The application will therefore be designed around:

- **Visual guidance** through 3D views, icons and colour rather than dense text
- **Local languages**, starting with Hindi and English
- **Few inputs**, with the truck saved once and reused on every trip
- **Large, simple controls** that can be used at a loading point
- **Low data use**, so it remains usable on highways with weak coverage
- **Step-by-step instructions** that loaders can follow without training

3.3 Why a solution is required

Existing tools address parts of the problem in isolation (Section 4.3). Load-planning software is built for warehouses and large shippers, freight marketplaces list loads without considering how they fit into a truck that is already partly loaded, and navigation apps do not account for cargo or the realities of freight travel. Nothing currently brings loading, additional cargo and route planning together for small operators.

4. Why It Matters

~75%
of India's trucking market is run by operators with up to five trucks[1]

28–43%
of truck travel is estimated to be empty running[1]

53%
of surveyed drivers earn ₹10,000–20,000 a month[3]

~12 hrs
of driving a day on average, covering about 417 km[3]

4.1 Economic impact

India spent an estimated **7.97% of its GDP, about ₹24 lakh crore, on logistics in 2023–24**, with fuel, congestion and heavy reliance on trucking identified among the main cost drivers.[2] When trucks run partly empty, the cost of each tonne moved rises, and that cost is passed on through the supply chain. Because most trucks belong to small operators, improving how their capacity is used has an impact across the entire sector, not only within a few large companies.

4.2 Drivers' earnings and living conditions

A nationwide study of more than 1,200 truck drivers found that most earn between ₹10,000 and ₹20,000 a month, drive around 12 hours a day, and that 93% receive no social security beyond their salary. Almost half reported driving while fatigued, and 84% said they would not recommend the profession to their own family.[3] These conditions are closely tied to how trips are planned: a half-empty truck earns less for the same hours, and an unplanned journey leaves drivers with long stretches between proper rest stops.

Condition drivers face	How Spareway can help
Low and uncertain earnings	More paid cargo on each trip raises trip income. For owner-drivers, this adds directly to their earnings; for fleets, it could support better pay and incentives, depending on how owners choose to distribute the additional earnings.
Long hours and fatigue	Routes are planned with suitable dhabas, hotels and rest points, so halts can be taken at the right places rather than wherever fatigue sets in.
Physical strain of handling	A clear loading sequence avoids unloading and reloading cargo at intermediate stops.
Penalties and informal payments	Weight and dimension checks keep loads within limits, reducing the risk of fines and roadside demands.

4.3 Existing solutions and their limitations

Solution	What it offers	Limitations for our users
LoadOptimizer.ai[4]	3D load plans for trucks, containers and pallets, with weight, stacking and orientation rules	Aimed at manufacturers, freight forwarders and 3PL providers; requires item lists in CSV or Excel; paid plans priced in euros; no camera-based scanning, no route planning and no search for additional cargo
Vahak[5]	Online marketplace to find loads and book trucks, with tracking and a freight rate calculator	Lists loads but does not show whether a load fits into a partly loaded truck; no load arrangement, space utilisation or route planning described
BlackBuck[6]	Large digital loads marketplace, FASTag, GPS tracking, fuel cards and vehicle loans	Strong on payments and tracking, but does not describe load arrangement, capacity planning, or route planning around tolls, checkpoints and rest stops
General navigation apps	Turn-by-turn directions, travel times and places of interest	Designed for general travel; not connected to the cargo on board, the remaining capacity or the cargo available along the route

Based on each company's public website and product description, reviewed on 28 October 2026. Where a feature is described as absent, it was not found in those descriptions; this does not mean the product cannot provide it.

Each of these products handles one part of the problem well. Spareway aims to bring together the three decisions that determine how well a truck is used: how cargo is loaded, which additional cargo can be carried and which route to take. What sets it apart is that each additional load and each route option is judged by its **net benefit to the operator**, after accounting for the extra costs it brings.

5. Proposed Solution Direction

We propose **Spareway**, a mobile-first web application that helps truck operators use their vehicles to the maximum feasible capacity across the entire trip. It supports the operator from the moment loading begins to the final drop: scanning the cargo, planning its arrangement in 3D, guiding the loaders, identifying additional cargo along the route, and recommending routes that are fast, affordable and practical. Every recommendation respects the truck's physical limits, its payload and the order of its stops, and is assessed on its net benefit to the operator.

01 **Before loading**	02 **During loading**	03 **On the route**
› Truck profile saved once › Cargo scanned with the phone camera › 3D loading plan generated	› Step-by-step guided loading › Live space and weight remaining › Plan adjusted if cargo changes	› Additional cargo along the corridor › New cargo fitted into the 3D plan › Route, halts and costs optimised
One trip record links the truck, its cargo, its loading plan and its route		

5.1 Core features

1 **Truck profile**

The operator records the truck's internal dimensions, payload limit, door position and running cost per kilometre once. This profile is reused on every trip, so the planning on each journey begins from accurate vehicle data rather than estimates.

2 **Guided cargo scanning**

Instead of typing out measurements, the user scans the cargo with the phone camera. The application guides them to move around the goods and capture them from several angles. The application is intended to detect individual items, estimate their dimensions using a reference object of known size, and identify characteristics such as fragile or upright-only goods. The user confirms or corrects the results and enters weights from the consignment documents.

3 **3D loading planner**

Using the truck profile and the scanned cargo, Spareway calculates an arrangement that makes the fullest feasible use of the space. The plan keeps the total within the payload limit, places heavier items lower, supports every stack fully, protects fragile goods and places cargo for later stops deeper inside the truck. The arrangement is shown as an interactive 3D model, coloured by destination, with the remaining space and payload displayed at all times.

4 **Guided loading from the start**

The 3D plan is used from the first item loaded. Loaders follow a numbered sequence that shows which item goes where, one step at a time. If the cargo that arrives differs from what was scanned, the plan is recalculated and the guidance updated, so that the truck is always loaded according to a valid arrangement.

5 Additional cargo along the route

Spareway will identify consignments whose pickup and delivery points lie along or near the truck's planned corridor. Each opportunity is first checked for feasibility against the space and payload still available and the existing delivery sequence, and its position is shown within the current 3D loading plan. Feasible opportunities are then assessed on their **net benefit**: the expected freight revenue minus the incremental costs the load introduces, including additional distance, fuel, tolls, handling, waiting time and any delay to existing deliveries.

> **Net benefit** = expected freight revenue − (additional distance cost + tolls + handling + waiting and delay costs)

A load is recommended only when it is feasible and its expected additional revenue justifies these incremental costs. Costs the truck would incur on its planned trip in any case are not counted again. Where a load falls short, the application indicates the minimum freight rate at which carrying it would become worthwhile.

6 Route and corridor optimisation

Spareway compares possible routes and corridors between the origin, each drop and any additional pickups. Routes are assessed on several parameters together: distance, expected traffic, toll charges, checkpoints and state borders, heavy-vehicle entry timings in cities, fuel stations, and the availability of dhabas, hotels and safe parking. Each option is compared on its overall trip cost and on the net earnings of the cargo it allows the truck to carry. The recommended route balances speed, cost and practicality, and it includes suggested halts so that drivers can rest at suitable places.

7 Trip cost and earnings summary

For every trip, Spareway brings together the cargo carried, capacity used, distance, fuel, tolls, revenue and net earnings. Over time, this record helps operators see how well their trucks are utilised and which routes and cargo are most profitable.

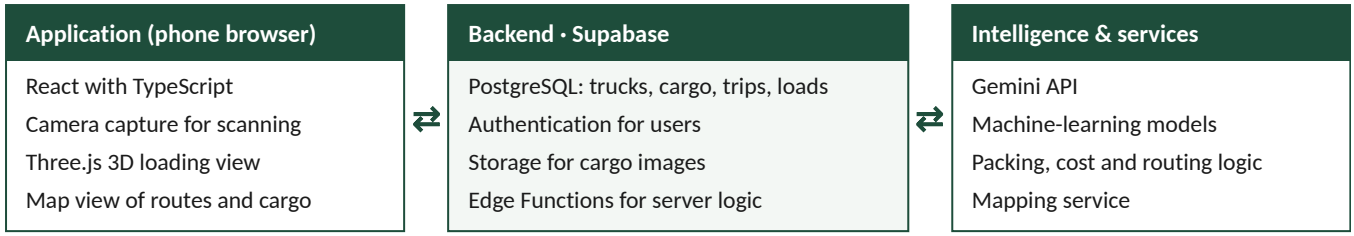
5.2 How every part of the problem is addressed

Problem identified	Spareway feature	Expected outcome
Unused cargo space on trips	3D loading planner; additional cargo along the route	Higher utilisation of space already paid for
Cargo arranged without a plan	Cargo scanning; guided loading	Tighter, safer arrangement from the first item
Overloading and unstable stacks	Payload and stacking checks in the planner	Loads kept within legal and safe limits
Cargo blocking earlier drops	Stop-order placement	Each stop unloaded without reshuffling
Extra cargo hard to find and evaluate	Corridor matching with feasibility and net-benefit checks	Only loads that remain profitable after incremental costs are recommended
Routes chosen by habit	Multi-parameter route evaluation on trip cost	Lower cost, fewer delays, planned rest
Low visibility of earnings	Trip cost and earnings summary	Better decisions on routes and cargo

6. Potential Technology Approach

Spareway is proposed as a **React-based progressive web application** designed for use on mobile phones and computers, with **Supabase** providing backend and data services. The architecture will connect cargo recording, 3D loading visualisation, capacity calculations, freight matching and route evaluation through a shared trip record.

Each technology will serve a defined purpose. AI may assist with cargo identification and recommendation explanations, while deterministic algorithms will handle calculations and constraints that require consistent results, including payload limits and geometric placement. Mapping and routing services will provide route information according to their available coverage and capabilities.



Component	Technology	Role in the system
Application	React, TypeScript, Tailwind CSS; progressive web app	A single interface that works on low-cost Android phones and computers without installation from an app store.
Backend and data	Supabase (PostgreSQL, Auth, Storage, Edge Functions)	Stores truck profiles, cargo, trips, load listings and images in a shared trip record; handles user accounts; runs server-side functions that call external services securely.
Cargo scanning	Device camera; reference-object measurement; object-detection model (TensorFlow.js or ONNX)	Guides the user through multi-angle capture and estimates item dimensions against a reference object of known size. A trained detection model for recognising individual items is a later step.
AI assistance	Gemini API	Assists with identifying cargo types and handling needs from images, and explains recommendations in plain Hindi or English.
Validation models	Separate machine-learning models	Cross-check AI outputs, for example comparing item counts and sizes with the detection model, before results are used.
Packing engine	TypeScript algorithm	Calculates placements under fixed rules for space, payload, support, fragility and stop order. The same inputs always produce the same arrangement.
Cost and net-benefit logic	TypeScript algorithm	Calculates the incremental cost and net benefit of each additional load and the overall cost of each route option.
Freight matching	Supabase queries and matching logic	Finds listed consignments along the planned corridor and passes them to the feasibility and net-benefit checks.
3D loading view	Three.js	Displays the exact positions calculated by the packing engine and the step-by-step loading sequence.
Maps and routing	Mappls or OpenStreetMap with an open routing engine (OSRM or GraphHopper)	Generates route options according to the coverage of the chosen service. Tolls, checkpoints, dhabas and hotels along the routes covered will be stored as a prepared dataset; real-time traffic will be used only where the service provides it.

The hackathon prototype will demonstrate the core workflow using sample and synthetic data. Features requiring reliable live freight listings, specialised computer-vision models, real-time traffic feeds or external integrations will be identified according to their implementation status rather than presented as completed capabilities.

7. Team Members, Roles & Responsibilities

Name	Role	Responsibilities
Agresh Agrawal	Team Leader, Technical Lead	Leads the team and the technical direction: system architecture, the packing and routing logic, Supabase and Gemini integration, and bringing all modules together.
Bhavya Agarwal	Developer	Builds the application screens, cargo scanning flow, 3D loading view and map interface, and tests each module.
Vansh Agarwal	Ideation and Planning	Shapes the idea, leads research with operators and drivers, prepares route and cargo data, and keeps the build on schedule.
Yash Agarwal	Design and Visuals	Designs a simple, driver-friendly interface, and prepares diagrams, presentation visuals and the final demonstration.

Sources

1. NITI Aayog, RMI and RMI India, *Fast Tracking Freight in India: A Roadmap for Clean and Cost-Effective Goods Transport*, June 2021, pp. 60–61.
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